Exploring Excess Return Generation through European Fixed Income Arbitrages

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Duarte et al. (2007) pose the question that motivates this paper:

> *Do fixed-income arbitrage strategies earn genuine alpha, or are their returns simply compensation for systematic risk — "picking up nickels in front of a steamroller"?*

Their answer: the more complex strategies generate **significant alpha** that survives a 10-factor benchmark in U.S. markets.

Open questions that motivate this paper:

- Does the result generalise to **European** fixed-income markets?
- Does it survive a **much more demanding** battery of factor controls?
- And — perhaps above all — **why does the alpha persist**?

Research Questions

RQ1. *Does alpha exist?*

Do three European fixed-income arbitrage strategies — the BTP Italia inflation-linked basis, the corporate CDS–Bond basis, and the iTraxx index skew — generate excess returns that survive standard benchmark factor models?

→ Section 3

RQ2. *Does alpha survive aggressive risk adjustment — and is it state-dependent?*

Does the alpha survive when the candidate factor universe is enlarged to **85** controls and selected by data-driven methods (rolling PCA; Adaptive Elastic Net with stability selection)? And: does the residual alpha vary systematically with intermediary funding stress?

→ Section 4

RQ3. *Why does alpha persist?*

Are the three mispricings linked to one another, and is the co-movement consistent with the slow-moving capital framework of Duffie (2010)?

→ Section 5

We extend Duarte et al. (2007) in four respects:

1. **New strategies, never studied in this combination.** To our knowledge, BTP Italia inflation-linked basis, European CDS–Bond basis in iTraxx Main, and iTraxx index skew across the four sub-indices have never been studied jointly as systematic arbitrages.
2. **Strategies with deterministic terminal payoff.** Unlike Duarte's swap-spread, yield-curve, volatility-selling and capital-structure trades, each of our three converges to a *certain* profit if held to maturity.
3. **Far more demanding factor controls.** Three classical benchmarks (Duarte et al. (2007), Brooks et al. (2020), Fung and Hsieh (2004)), *plus* rolling PCA on 85 candidate factors (Ludvigson and Ng, 2009), *plus* Adaptive Elastic Net (Zou and Zhang, 2009) with stability selection (Meinshausen and Bühlmann, 2010).
4. **We test *why* the alpha persists.** We map our findings to the predictions of the slow-moving capital literature (Duffie (2010), Brunnermeier and Pedersen (2009), Mitchell and Pulvino (2012), Boyarchenko et al. (2016)), and exploit the European setting as a natural experiment on investor-base segmentation.

Basis definition (following Fleckenstein et al. (2014), adapted to the Italian instrument):

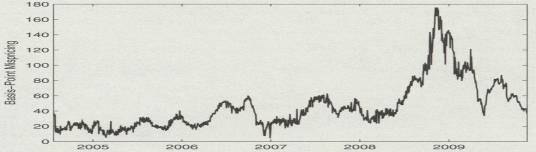
$$m_t^{\text{BTPi}} = \underbrace{y_t^{\text{BTPi}} + \pi_t^{\text{swap}}}_{\text{synthetic nominal yield}} - y_t^{\text{nominal BTP}}$$

Why this differs from TIPS:

- **No ballooning risk.** Inflation revaluation paid out semiannually $\Rightarrow$ inflation factor resets to unity at each coupon $\Rightarrow$ no contamination from default exposure on accumulated revaluation.
- **Retail bondholder base.** Primary placement reserved to households + *premio fedeltà* for buy-and-hold $\Rightarrow$ preferred-habitat investors with no leverage, no margin calls, no forced unwinding.

Consequence: the BTP Italia basis behaves differently from TIPS during stress—and persists.

TIPS basis pre-2010 (Fleckenstein 2014):



TIPS basis post-2010: has vanished

Strategy 2: CDS–Bond Basis — The Negative Basis Trade

Basis definition:

$$m_t^{\text{CDS–Bond}} = \mathrm{CDS}_t^{\text{spread}} - z_t^{\text{spread}}$$

where z_t^{spread} is the bond's z-spread over the euro zero-coupon swap curve. A *negative* basis means credit protection via CDS is cheaper than the bond-implied credit spread.

The trade:

- Buy the corporate bond (earn z-spread); buy CDS protection (pay lower CDS premium).
- **Locked-in positive carry** until convergence or bond maturity.
- In default: bond recovers RR on par; CDS pays $100 - RR \Rightarrow$ **combined position returns par regardless of recovery** (provided CDS tenor $\geq$ bond residual life).

Universe: on-the-run iTraxx Europe Main 5Y constituents (investment-grade, liquid). All CDS centrally cleared (LCH/ICE) $\Rightarrow$ no counterparty credit risk on the protection leg.

Restriction to negative basis: the reverse trade requires reverse-repoing the corporate bond, subject to recall risk in stress (Bai and Collin-Dufresne, 2019).

Basis definition:

$$\mathrm{Skew}_t \;=\; S_t^{\text{index, spread}} \;-\; \sum_{i=1}^{N} w_{i,t}\, S_{i,t}^{\text{single, spread}}$$

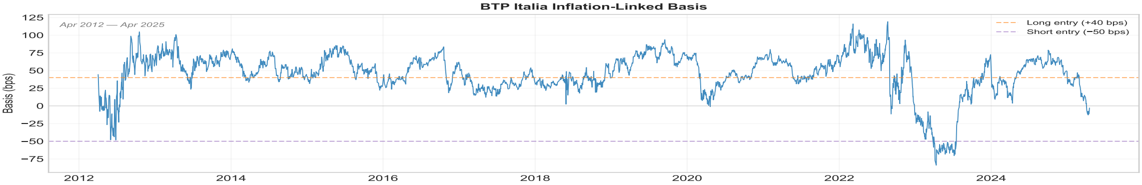
Quoted spreads in bps p.a.; $w_{i,t}$ are duration weights. A positive (negative) skew indicates that the index trades *wide* (*tight*) relative to the sum of its parts.

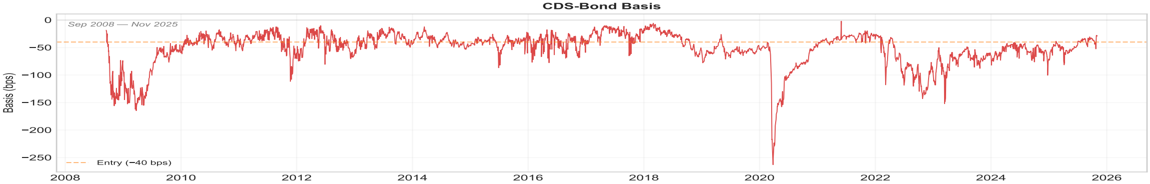
Why a non-zero skew can persist:

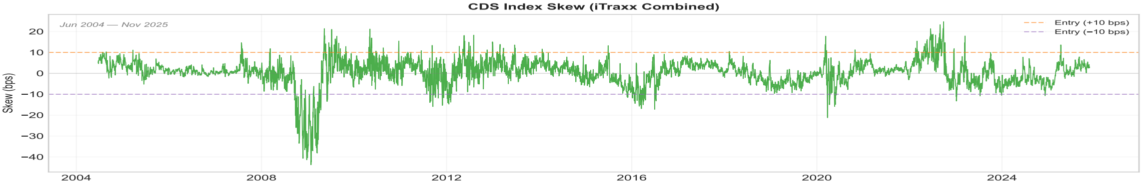
- **Liquidity differential.** Single-name CDS are far less liquid than the on-the-run index, with wider bid-ask spreads and shallower order books, especially in stress (Junge and Trolle, 2015). The skew is well documented as a market-wide liquidity proxy.
- **Strategic positioning by institutional investors.** Index CDS attract concentrated trading demand for macro credit hedging, making the index leg structurally tighter than the basket of single names that replicates it (Boyarchenko et al., 2016).
- **Execution complexity of the replicating basket.** Replicating the index requires entering $\sim$125 single-name positions versus a single index contract, with bid-ask costs and margin spread across many bilateral lines at trade inception and exit.

The trade: go long/short the index versus the duration-weighted single-name basket, depending on the sign of the skew. Entered on the on-the-run series and held to convergence (no rolling). Both legs centrally cleared.

Market-Level Basis Time Series







Trade Construction & Index Aggregation

Following Duarte et al. (2007). Each individual trade is treated as a separate fictional fund (own entry/exit/return stream); the strategy index on any day is the weighted average of the returns of the funds active on that day.

Two weighting schemes:

Equal-Weighted (Duarte et al., 2007)

$$r_t^{\mathrm{EW}} = \frac{1}{K_t} \sum_{k=1}^{K_t} r_{k,t}$$

Signal-Weighted (Rebonato and Ronzani, 2021)

$$r_t^{\mathrm{SW}} = \sum_{k=1}^{K_t} \frac{|b_{k,0} - \theta_{t_k}|}{\sum_{j=1}^{K_t} |b_{j,0} - \theta_{t_j}|} r_{k,t}$$

$b_{k,0}$: basis at entry of trade k; θ_{t_k}: expanding-window historical equilibrium up to entry. Weight $\propto$ distance from equilibrium $\rightarrow$ richer dislocation, larger allocation.

Implementation choices.

- **SW dominates EW** on Sharpe and MPPM for all three strategies $\rightarrow$ SW baseline; EW in appendix confirms robustness.
- Three-regime classification on iTraxx Main 5Y — *LOW* ($<$ 60 bps), *MID* (60–100 bps), *HIGH* ($>$ 100 bps) — drives **regime-dependent transaction costs** (entry/exit fees proportional to DV01).

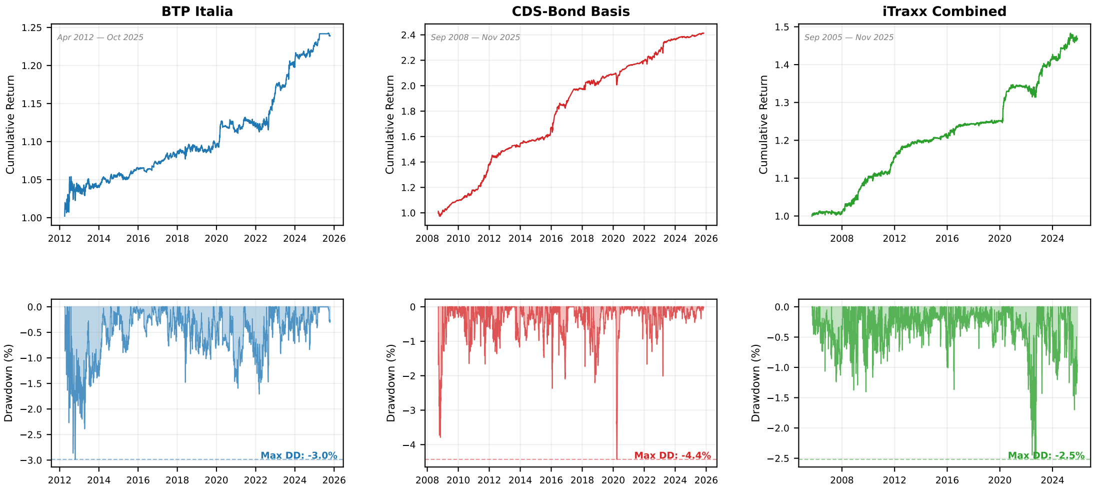
Strategy Performance: Summary Statistics

Annualised, daily data, net of regime-dependent transaction costs.

	BTP Italia	*CDS–Bond Basis*	*iTraxx Combined*
Sample	Mar 2012 – May 2025	Sep 2008 – May 2025	Jun 2004 – May 2025
Mean return (%)	1.67	**5.37**	1.96
Volatility (%)	2.34	3.15	2.59
Sharpe ratio	0.71	**1.71**	0.76
Skewness	+0.43	+0.60	**+2.81**
Max drawdown (%)	3.0	4.4	2.5
Basis ρ_1 (daily)	0.98	0.92	0.87

- Sharpe ratios from **0.71** (BTP Italia) to **1.71** (CDS–Bond) — economically large.
- *Positive skewness* for all three (max +2.81 for iTraxx) $\Rightarrow$ cuts against "picking up nickels in front of a steamroller".
- Persistence $\rho_1 \in [0.87, 0.98]$ on the underlying basis $\Rightarrow$ slow-moving capital signature (Duffie, 2010).

Strategy Performance: Cumulative Returns and Drawdowns



MPPM and Volatility-Managed Performance

Manipulation-Proof Performance Measure (MPPM)

Strategy	MPPM (% p.a.)			Avg Return (% p.a.)	Vol (% p.a.)	Sharpe	N
	$\rho = 2$	$\rho = 3$	$\rho = 4$				
CDS-Bond Basis	5.06	5.01	4.97	5.16	3.07	1.68	207
iTraxx *combined*	1.83	1.82	1.80	1.85	1.52	1.22	243
BTP-Italia	1.56	1.55	1.54	1.59	1.62	0.98	163

MPPM as in Goetzmann et al. (2007), based on monthly returns and reported in annualized percentage terms. All moments are annualized.

Volatility-Managed Portfolios (Moreira–Muir, 2017)

	Buy-and-Hold		Vol-Managed		Regression			*N*
Strategy	Ret.	Sharpe	Ret.	Sharpe	α	β	R^2	
BTP Italia	1.52	0.95	0.37	0.23	-0.22	0.39	0.150	158
CDS–Bond Basis	5.09	1.70	3.49	1.16	+1.68	0.36	0.127	202
iTraxx Combined	1.91	1.25	0.84	0.55	-0.16	0.53	0.279	242

- All three strategies deliver positive MPPM, with the highest utility-based performance for the CDS–Bond basis and iTraxx strategies.
- Volatility timing generates a sizable premium for the CDS–Bond basis, while effects are more modest for BTP Italia and iTraxx, consistent with their lower return variance.

We evaluate our three fixed–income arbitrage strategies under the established risk–factor models used in the literature:

- *Duarte, Longstaff & Yu (2007), Review of Financial Studies*.
  Fixed–income arbitrage benchmark used to assess exposure to equity, credit and Treasury–based relative–value factors.
- *Fung & Hsieh (2004), Financial Analysts Journal*.
  Trend–following and option–based factors widely used to benchmark hedge–fund strategies.
- *Brooks, Gould & Richardson (2020), Journal of Fixed Income*.
  Active fixed–income premium decomposition: term, credit, EM, inflation–linked, and volatility premia.

US factors are mapped to their EUR counterparts. European equity factors use the Fama–French Europe datasets, converted from USD into EUR following the methodology of Glück, Hübel & Scholz (JPM, 2020). Trend–following factors (Fung–Hsieh) are global and directly applicable.

Duarte, Longstaff & Yu (2007) — 10-Factor FI Arbitrage Model

	α (% p.a.)	β_{Mkt-RF}	β_{SMB}	β_{HML}	β_{UMD}	β_{RS}	β_{RI}	β_{RB}	β_{R2}	β_{R5}	β_{R10}	R^2 adj	N
Panel A: US Factors													
BTP Italia	**2.05*****	-0.07***	-0.03**	-0.04**	0.00	0.04***	0.04	-0.08	0.09	-0.05	0.04	0.132	161
	(4.41)	(-3.40)	(-2.19)	(-2.44)	(0.07)	(3.40)	(0.55)	(-0.94)	(0.35)	(-0.31)	(0.54)		
CDS Bond Basis	**5.13*****	-0.01	0.00	0.01	-0.03*	-0.01	0.26***	-0.06	0.11	-0.46*	0.12	0.174	204
	(6.66)	(-0.43)	(0.08)	(0.40)	(-1.75)	(-0.39)	(3.32)	(-1.59)	(0.31)	(-1.84)	(1.08)		
iTraxx Indices Skew	**1.81*****	0.02	-0.02*	-0.02	-0.02*	-0.01	-0.02	-0.02	0.34	-0.14	0.05	0.051	240
	(4.64)	(1.55)	(-1.70)	(-1.28)	(-1.91)	(-1.44)	(-0.37)	(-0.71)	(1.53)	(-0.99)	(0.84)		
Panel B: EUR Factors													
BTP Italia	**1.65*****	-0.05**	-0.06***	-0.05**	0.02	0.02	-0.01	0.03	0.05	-0.03	-0.03	0.108	162
	(3.36)	(-1.99)	(-2.97)	(-2.05)	(0.99)	(1.27)	(-0.08)	(0.36)	(0.11)	(-0.14)	(-0.27)		
CDS Bond Basis	**5.14*****	0.00	0.02	0.04	-0.03	-0.03**	0.47***	-0.07*	-0.53	0.15	-0.17	0.144	205
	(5.59)	(0.08)	(0.53)	(1.13)	(-1.38)	(-2.01)	(3.87)	(-1.70)	(-0.80)	(0.32)	(-1.05)		
iTraxx Indices Skew	**2.26*****	-0.02	-0.04***	-0.07***	-0.03**	0.01	0.06	-0.02	0.02	0.06	-0.07	0.091	241
	(4.64)	(-1.20)	(-2.60)	(-2.58)	(-2.27)	(1.16)	(1.00)	(-0.99)	(0.05)	(0.26)	(-0.90)		

Fung & Hsieh (2004) — 7-Factor Hedge Fund Model

	α (%)	$\beta_{S\&P}$	β_{SC-LC}	β_{BdOpt}	β_{FXOpt}	β_{ComOpt}	β_{10Y}	$\beta_{CredSpr}$	R^2 adj	N
Panel A: US Factors										
BTP Italia	**1.80*****	-0.02*	-0.02**	0.12	-0.01	0.33	0.00	0.28	0.104	162
	(3.64)	(-1.96)	(-2.08)	(0.72)	(-0.03)	(1.05)	(0.01)	(1.18)		
CDS Bond Basis	**5.28*****	-0.02	0.01	0.09	-0.47	-0.09	-0.98***	-1.73***	0.185	205
	(6.68)	(-1.58)	(0.53)	(0.35)	(-1.49)	(-0.27)	(-3.98)	(-5.35)		
iTraxx Indices Skew	**1.69*****	0.02	-0.02**	0.10	0.17	0.13	-0.07	0.16	0.029	241
	(4.32)	(1.46)	(-2.09)	(0.48)	(0.89)	(0.48)	(-0.57)	(0.78)		
Panel B: EUR Factors										
BTP Italia	**1.86*****	-0.04***	-0.06**	0.34*	-0.08	0.34	0.28	-0.53	0.111	162
	(3.64)	(-2.99)	(-2.40)	(1.80)	(-0.39)	(0.92)	(1.44)	(-1.13)		
CDS Bond Basis	**4.67*****	-0.03	0.04	0.21	-0.71*	-0.59	-0.88**	-0.74	0.112	195
	(5.38)	(-1.60)	(1.22)	(0.88)	(-1.80)	(-1.64)	(-2.38)	(-1.32)		
iTraxx Indices Skew	**3.48*****	-0.00	-0.07**	0.12	-0.13	0.59*	-0.34	-0.12	0.051	195
	(6.05)	(-0.28)	(-2.02)	(0.53)	(-0.56)	(1.83)	(-1.37)	(-0.38)		

Brooks, Gould & Richardson (2020) — Active FI Premia

	α (% p.a.)	β_{Term}	$\beta_{GlobalTerm}$	$\beta_{GlobalAgg}$	$\beta_{Infl.Linkers}$	$\beta_{CorpCredit}$	$\beta_{EmergDebt}$	$\beta_{EmergCurr}$	β_{USTVol}	R^2 adj	N
Panel A: US Factors											
BTP Italia	**1.88*** **	0.15	-0.27	0.13	-0.02	-0.14**	0.09*	-0.02	0.00	0.115	163
	(3.65)	(0.98)	(-0.88)	(0.31)	(-0.82)	(-2.37)	(1.72)	(-0.91)	(0.15)		
CDS Bond Basis	**4.61*** **	0.01	-0.20	0.42	-0.03	-0.00	0.07	0.02	0.01	0.104	207
	(5.38)	(0.08)	(-0.43)	(0.73)	(-0.44)	(-0.02)	(0.97)	(0.43)	(0.40)		
iTraxx Indices Skew	**1.57*** **	-0.05	-0.10	0.21	-0.01	0.04	-0.07*	0.03	0.04**	0.098	243
	(4.36)	(-0.45)	(-0.43)	(0.68)	(-0.17)	(1.08)	(-1.82)	(1.49)	(2.00)		
Panel B: EUR Factors											
BTP Italia	**1.96*** **	-0.07	-0.36	0.51*	-0.07**	-0.06	-0.03	0.03	0.00	0.009	163
	(3.28)	(-1.47)	(-1.43)	(1.87)	(-2.01)	(-1.23)	(-0.93)	(1.01)	(0.21)		
CDS Bond Basis	**5.07*** **	-0.02	-0.05	0.25	-0.02	-0.08	0.15**	-0.02	-0.02	0.039	207
	(5.87)	(-0.17)	(-0.12)	(0.70)	(-0.24)	(-1.52)	(2.21)	(-0.31)	(-1.30)		
iTraxx Indices Skew	**1.96*** **	0.03	-0.24	0.26	-0.02	0.01	-0.03	-0.02	-0.02*	0.008	243
	(4.73)	(0.32)	(-1.09)	(1.34)	(-0.50)	(0.81)	(-1.30)	(-1.16)	(-1.76)		

Cross-Model Synthesis: Alpha is Robust and *State-Dependent*

Full-sample alpha (% p.a.)

	BTP Italia	*CDS–Bond*	*iTraxx*
Duarte et al. (2007)	$+1.65^{***}$	$+5.14^{***}$	$+2.26^{***}$
Brooks et al. (2020)	$+1.93^{***}$	$+5.07^{***}$	$+1.96^{***}$
Fung & Hsieh (2004)	$+1.86^{***}$	$+4.67^{***}$	$+3.48^{***}$

Conditional Alpha *(Ferson–Schadt, α_1 per 1σ stress)*

	BTP Italia	*CDS–Bond*	*iTraxx*
Duarte et al. (2007)	$+1.09^{*}$	$+2.87^{***}$	$+1.06^{***}$
Brooks et al. (2020)	$+1.50^{**}$	$+2.90^{***}$	$+1.26^{***}$
Fung & Hsieh (2004)	$+1.39^{*}$	$+3.31^{***}$	$+2.81^{***}$

Regime decomposition *(HIGH: iTraxx Main 5Y $>$ 100 bps)*

	Duarte et al. (2007)		Brooks et al. (2020)		Fung & Hsieh (2004)	
	NORM	HIGH	NORM	HIGH	NORM	HIGH
BTP Italia	$+1.65^{***}$	$+4.74^{**}$	$+1.42^{***}$	$+4.13^{**}$	$+1.46^{***}$	$+5.62^{***}$
CDS–Bond Basis	$+3.77^{***}$	$+8.63^{***}$	$+3.63^{***}$	$+9.69^{***}$	$+2.95^{***}$	$+9.56^{***}$
iTraxx Combined	$+1.98^{***}$	$+2.71^{***}$	$+1.38^{***}$	$+4.38^{***}$	$+2.61^{***}$	$+7.25^{***}$

- *Full-sample*: alpha consistent across the three frameworks.
- *Regime decomposition*: HIGH-stress alpha is **2–4×** NORMAL alpha — in every cell.
- *Conditional alpha*: $\alpha_1 > 0$ everywhere; sig. at 1% for the institutional strategies ⇒ **slow-moving capital signature**.

The three benchmarks of Section 3:

- Alpha positive and significant across all 9 cells
- $\bar{R}^2 \in [1\%, 18\%]$ — bulk of return variation *unexplained*

Two open questions:

1. Are there *other* risk factors — outside the standard benchmarks — that span these returns?
2. If alpha survives even with a much richer factor universe, the case for genuine mispricing strengthens.

Strategy:

- Assemble **85 candidate factors** from the empirical asset pricing, intermediary, and macro-finance literatures.
- Two complementary approaches to exploit the broader factor universe: **rolling PCA** (Ludvigson and Ng, 2009) and **Adaptive Elastic Net** with stability selection (Zou and Zhang, 2009; Meinshausen and Bühlmann, 2010).

Factor universe (organized in 6 families)

- *Credit risk* (e.g., CDS indices, excess bond premium, sovereign spreads)
- *Liquidity & funding stress* (e.g., OIS spreads, repo indicators, noise/illiquidity measures)
- *Equity / intermediary proxies* (e.g., BAB, value, momentum)
- *Volatility* (e.g., VIX/V2X, MOVE, option-implied measures, trend-following)
- *Macro uncertainty & policy* (e.g., uncertainty indices, EPU, financial conditions)
- *Rates & term structure* (e.g., term, slope, swap spreads)

$2^{85} \approx 3.87 \times 10^{25}$ subsets $\Rightarrow$ exhaustive search infeasible $\Rightarrow$ penalised methods needed.

Setup (Ludvigson and Ng, 2009):

- Rolling PCA, window $W = 24$ months, $K = 8$ components. Across rolling windows, the eight components explain 80.8% of variance on average (75.7%–87.8%).
- Standardisation *within* each estimation window $\Rightarrow$ no look-ahead.
- Both contemporaneous ($PC_t \rightarrow r_t$) and predictive ($PC_t \rightarrow r_{t+1}$) timing.

	Contemporaneous		Predictive	
Strategy	α (% p.a.)	$\bar{R}^2$	α (% p.a.)	$\bar{R}^2$
BTP Italia	$+1.61^{***}$ (3.48)	0.054	$+1.68^{***}$ (4.05)	-0.005
CDS–Bond Basis	$+5.87^{***}$ (5.74)	0.120	$+5.43^{***}$ (5.91)	0.050
iTraxx Combined	$+2.21^{***}$ (5.01)	0.049	$+2.07^{***}$ (5.38)	0.041

- **Alpha survives at 1%** for all three strategies, both timings.
- $\bar{R}^2$ stays low (0–12%) $\Rightarrow$ we turn to **machine-learning** methods (AEN) to identify factors that better span these returns.

Two-stage penalised regression of Zou and Zhang (2009):

$$\hat{\beta}^{\text{AEN}} = \arg\min_{\beta} \left\{ \frac{1}{T} \sum_t \left(r_t - \alpha - \beta' f_t \right)^2 + \lambda_2 \|\beta\|_2^2 + \lambda_1 \sum_{j=1}^{p} w_j |\beta_j| \right\}, \quad w_j = |\hat{\beta}_j^{\text{EN}}|^{-\gamma},\ \gamma = 1.$$

- **Adaptive weights** (Stage 1 EN $\rightarrow$ Stage 2 weights): factors with large Stage-1 coefficients are penalised *less*; near-zero ones are penalised *more* $\Rightarrow$ restores the *oracle property* (LASSO is inconsistent under correlated factors).
- ℓ_2 **ridge** $\Rightarrow$ *grouping effect*: highly correlated factors selected/dropped *together*, not arbitrarily.
- **Tuning** (λ_1, λ_2) via Hannan–Quinn (HQC); coordinate descent (Friedman et al., 2010).
- **Stability selection** (Meinshausen and Bühlmann, 2010): $B = 100$ stationary bootstrap (expected block 9m), retain factor only if selection frequency $\hat{\pi}_j \geq 80\%$ $\Rightarrow$ formal guarantee on expected false selections.
- **Post-selection OLS** on stable set, with Newey–West HAC $\Rightarrow$ unbiased inference + sparsity.

AEN Stable-Factor Model: BTP Italia

Variable	Coefficient	t-stat	p-value
α	+1.39***	3.34	0.00
$\mathrm{SMB_{EU}}$	-0.04**	-2.30	0.02
ILLIQ	+0.64**	2.50	0.01
SS10Y	-0.02***	-3.26	0.00
$\mathrm{UMD_{EU}}$	+0.03**	1.96	0.05
$\mathrm{CDX_{IG}}$	+0.01	1.60	0.11
$\mathrm{LIBOR_{OIS}}$	+0.49*	1.90	0.06

Model fit

T (obs)	157
k (factors)	6
R^2	0.2474
R^2_{adj}	0.2173
Durbin–Watson	2.0351

(α) annualized %.
*** $p<1\%$, ** $p<5\%$,
* $p<10\%$.

- $\mathrm{SMB_{EU}}$: European equity size factor (small − big caps). *Fama and French (2017)*
- ILLIQ: Change in the Amihud illiquidity shock measure. *Amihud and Mendelson (2015)*
- SS10Y: 10Y EUR swap spread (par swap rate − 10Y Bund yield). *Collin-Dufresne et al. (2001)*
- $\mathrm{UMD_{EU}}$: European equity momentum factor (winners − losers). *Carhart (1997)*
- $\mathrm{CDX_{IG}}$: First difference of the CDX Investment Grade index CDS spread. *Klaus and Rzepkowski (2009)*
- $\mathrm{LIBOR_{OIS}}$: Libor–OIS spread (3M LIBOR − 3M OIS). *Nyborg and Ostberg (2014)*

AEN Stable-Factor Model: CDS–Bond Basis

Variable	Coefficient	*t*-stat	*p*-value
α	+3.94***	3.66	0.00
$\Delta\mathrm{UF}$	-4.27***	-2.61	0.01
$\mathrm{RI}_{\mathrm{EU}}$	+0.16***	3.27	0.00
$\mathrm{EMERG}_{\mathrm{FX}}$	+0.06**	1.99	0.05
$\mathrm{PB}_{\mathrm{EU\_CDS\_1Y}}$	+0.01**	2.02	0.04
SS5Y	+0.02**	2.33	0.02
$\mathrm{CRED}_{\mathrm{SPR\_US}}$	-0.51**	-2.20	0.03
$\mathrm{EP}_{\mathrm{SVIX\_1M}}$	+3.25***	2.82	0.00
PTFSFX	-0.45*	-1.74	0.08

Model fit

T (obs)	197
k (factors)	8
R^2	0.2969
R^2_{adj}	0.2670
Durbin–Watson	2.0233

(α) annualized %.
*** $p<1\%$, ** $p<5\%$,
* $p<10\%$.

- $\Delta\mathrm{UF}$: Change in financial uncertainty index. *Ludvigson et al. (2021)*
- $\mathrm{RI}_{\mathrm{EU}}$: Euro investment-grade industrial bond return factor. *Collin-Dufresne et al. (2001)*
- $\mathrm{EMERG}_{\mathrm{FX}}$: Equal-weighted basket of emerging market currencies vs USD. *Brooks et al. (2020)*
- $\mathrm{PB}_{\mathrm{EU\_CDS\_1Y}}$: European prime broker 1Y CDS spread (counterparty risk). *Klaus and Rzepkowski (2009)*
- SS5Y: 5Y EUR swap spread (par swap rate − 5Y Bobl yield). *Collin-Dufresne et al. (2001)*
- $\mathrm{CRED}_{\mathrm{SPR\_US}}$: US credit spread factor (Moody's BAA − AAA yield). *Fama and French (1989)*
- $\mathrm{EP}_{\mathrm{SVIX\_1M}}$: Option-implied equity premium via SVIX, 1-month horizon. *Martin (2017)*
- PTFSFX: Currency trend-following factor (lookback straddles on FX futures). *Fung and Hsieh (2004)*

AEN Stable-Factor Model: iTraxx Combined

Variable	Coefficient	t-stat	p-value
α	+0.47	1.04	0.30
$\mathrm{EP}_{\mathrm{SVIX\_1M}}$	+3.31***	2.89	0.00
$\mathrm{LIBOR}_{\mathrm{OIS}}$	+0.48**	2.21	0.03
$\Delta\mathrm{V2X}$	-0.02**	-2.55	0.01
$\Delta\mathrm{UR}$	+3.01**	2.35	0.02
$\Delta\mathrm{FAILS}_{\mathrm{PCT\_TSY}}$	+0.40***	3.20	0.00

Model fit

T (obs)	237
k (factors)	5
R^2	0.2521
R^2_{adj}	0.2359
Durbin–Watson	2.2489

(α) annualized %.
*** $p<1\%$, ** $p<5\%$,
* $p<10\%$.

- $\mathrm{EP}_{\mathrm{SVIX\_1M}}$: Option-implied equity premium via SVIX, 1-month horizon. *Martin (2017)*
- $\mathrm{LIBOR}_{\mathrm{OIS}}$: Libor–OIS spread (3M LIBOR − 3M OIS). *Nyborg and Ostberg (2014)*
- $\Delta\mathrm{V2X}$: Change in V2X (30-day implied vol on Euro STOXX 50). *Chung et al. (2019)*
- $\Delta\mathrm{UR}$: Change in real-activity uncertainty index. *Ludvigson et al. (2021)*
- $\Delta\mathrm{FAILS}_{\mathrm{PCT\_TSY}}$: Change in U.S. Treasury fails-to-deliver / total debt outstanding. *Fleckenstein et al. (2014)*

Cross-Method Comparison: Benchmarks vs PCA vs AEN

Method	*BTP Italia*		*CDS–Bond*		*iTraxx*	
	α (%)	$\bar{R}^2$	α (%)	$\bar{R}^2$	α (%)	$\bar{R}^2$
Duarte (k=10)	+1.65***	0.108	+5.14***	0.144	+2.26***	0.091
Brooks (k=8)	+1.96***	0.009	+5.07***	0.039	+1.96***	0.008
Fung–Hsieh (k=7)	+1.86***	0.111	+4.67***	0.112	+3.48***	0.051
PCA (K=8)	+1.61***	0.054	+5.87***	0.120	+2.21***	0.049
AEN (k=6/8/5)	+1.39***	**0.217**	+3.94***	**0.267**	+0.47	**0.236**

- **$\bar{R}^2$ jumps from 5–15% (benchmarks) to $\sim$ 22–27% (AEN)**: the AEN-selected factors absorb a much larger share of strategy return variation than any classical benchmark.
- **Alpha survives PCA and AEN for BTP Italia and CDS–Bond Basis**: a residual premium remains. **For iTraxx Combined alpha collapses** ($t \approx 1.04$): the skew premium is fully absorbed by AEN factors.

The AEN-selected factor sets are **economically interpretable** and **differ sharply across strategies**, mapping onto the economic structure of each trade.

1. **BTP Italia** → *liquidity, funding, and term-structure controls.*
   The stable set is dominated by the Amihud illiquidity measure, the Libor–OIS spread, the 10Y EUR swap spread, the European size and momentum factors — a combination of aggregate illiquidity, short-term funding stress, and the term structure of euro rates.

2. **CDS–Bond Basis** → *the classical intermediary asset-pricing channel.*
   The stable set is dominated by dealer balance-sheet and funding variables: the European prime-broker CDS spread, the Ludvigson financial uncertainty index, the SVIX equity premium proxy, the 5Y EUR swap spread, and the credit spread factor — exactly the channel that the slow-moving capital framework would predict for a trade requiring continuous repo financing of the cash bond (Boyarchenko et al., 2016).

3. **iTraxx Combined** → *aggregate funding stress and market-wide volatility.*
   The stable set is more macro-flavoured — the SVIX equity premium, Libor–OIS, V_2X innovations, real-activity uncertainty, and Treasury fails — consistent with the long-standing characterisation of the index skew as a market-wide liquidity proxy (Junge and Trolle, 2015). The collapse of the

What we have established:

- Alpha exists, robust to benchmarks, PCA on 85 factors, AEN with stability selection.
- Alpha rises in stress (regime decomposition, Ferson–Schadt, rolling).

The deeper question:
Why does alpha persist, and are the three mispricings linked?

Theoretical anchors:

- Duffie (2010) — qualitative framework of slow-moving capital.
- Brunnermeier and Pedersen (2009) — formal model with Proposition 6 on commonality of liquidity.
- Mitchell and Pulvino (2012) — empirical cross-strategy mechanism (*several months* timescale).
- Boyarchenko et al. (2016) — post-crisis equilibrium and funding-dependence.
- Fleckenstein et al. (2014) — cross-arbitrage spanning methodology.

	Prediction
P1	Mispricings serviced by a common pool of intermediary capital should *co-move*.
P2	The co-movement should *intensify* when intermediary capital is scarce.
P3	Mispricing changes in one strategy should *predict* those of others, propagating along the liquidity gradient.
P4	The *persistence* of mispricing levels should reflect the funding-dependence of each arbitrage trade.
P5	A common factor extracted from the co-movement should be explained by intermediary capital, *not* macro fundamentals.

Arbitrageur (all three strategies): Institutional — hedge funds, dealer prop desks, leveraged accounts.

Institutional bondholder base

CDS–Bond Basis
iTraxx Combined

Counterparty subject to margin calls, VaR limits, balance-sheet constraints.

VS

Retail bondholder base

BTP Italia

Households (loyalty bonus, inflation hedge for IT investors). No leverage, no margin calls, buy-and-hold.

Why this matters (Brunnermeier and Pedersen, 2009 + Vayanos and Vila, 2021):

- **Institutional bondholder pair**: stress → VaR spikes → risk budget exhausted → *both* bondholders + arbitrageurs unwind.
- **Retail bondholder**: insulated from funding shocks → no forced selling → basis remains open for arbitrageurs who can still deploy capital.

Natural experiment: *if co-movement were driven by macro, all three would move in the same direction regardless of who holds the bond.*

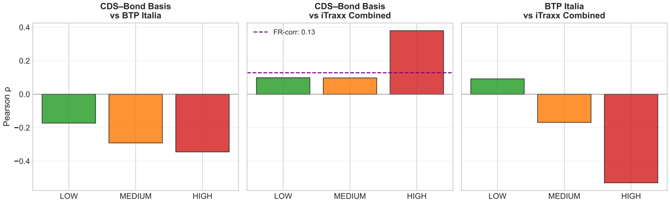
Pearson correlations of mispricing changes $\Delta m_{i,t}$, unconditional. HAC inference.

Pair	Pearson ρ	t-HAC	p-value	N
CDS–Bond Basis vs BTP Italia	$-0.260***$	-2.89	0.004	206
CDS–Bond Basis vs iTraxx Combined	$+0.182*$	1.65	0.098	206
BTP Italia vs iTraxx Combined	$-0.152**$	-1.99	0.047	156

- **Within institutional pair** (CDS–Bond $\leftrightarrow$ iTraxx): positive correlation $\Rightarrow$ consistent with a common pool of dealer capital.
- **Across investor bases** (BTP Italia pairs): negative correlation $\Rightarrow$ first signal of segmentation.
- Foundational anchor: Duffie (2010) catastrophe-insurance analogy + Brunnermeier and Pedersen (2009) commonality of liquidity (Prop. 6(i)).

P2: Stress Amplification

Regime correlations (Fig. 14):



Interaction $\hat{\beta}_1$:

$$\Delta m_i = \beta_0 \Delta m_j + \beta_1(\Delta m_j \times z_t) + \gamma z_t + \varepsilon$$

Direction	$\hat{\beta}_2$
BTP → CDS–Bond Basis	-0.0606
CDS–Bond Basis → BTP	-0.1557
iTraxx → CDS–Bond Basis	$+0.1862$
CDS–Bond Basis → iTraxx	$+0.0226$
iTraxx → BTP	$-1.2648**$
BTP → iTraxx	$-0.0275***$

- **Institutional pair (CDS–Bond ↔ iTraxx)**: ρ rises monotonically LOW → HIGH ⇒ *commonality of fragility* (Brunnermeier and Pedersen, 2009, Prop. 6(ii)).
- **Cross-base pair (BTP ↔ inst.)**: sign reversal in HIGH ⇒ *flight to quality* (Brunnermeier and Pedersen, 2009, Prop. 6(iv)).
- Interaction β_1 confirms continuous stress amplification with sign asymmetry.

$\Delta m_{i,t} = \alpha + \sum_{j \neq i} \sum_{l=0}^{2} \beta_{j,l} \, \Delta m_{j,t-l} + \varepsilon_t$ — *design of Fleckenstein et al. (2014). Newey–West HAC.*

Cross-strategy spanning ($\bar{R}^2$):

Dependent: $\Delta m_{i,t}$	$\bar{R}^2$
CDS–Bond Basis	9.7%
BTP Italia	13.3%
iTraxx Combined	5.4%

Granger causality (VAR on Δm):

Direction	p-value
$\rightarrow$ CDS–Bond Basis	0.370
$\rightarrow$ CDS–Bond Basis	0.127
$\rightarrow$ BTP Italia	0.047**
$\rightarrow$ BTP Italia	0.002***
$\rightarrow$ iTraxx Combined	0.892
$\rightarrow$ iTraxx Combined	0.077*

- **Hierarchy**: iTraxx (most liquid) $\rightarrow$ CDS–Bond $\rightarrow$ BTP Italia.
- BTP Italia is the **most predictable** ($\bar{R}^2$ highest): retail-held basis adjusts *last*, with 1–2 months delay.
- Consistent with funding-liquidity feedback of Brunnermeier and Pedersen (2009) and cross-strategy capital reallocation of Mitchell and Pulvino (2012).

P4: Persistence Reflects Funding-Dependence

$m_{i,t} = c_i + \phi_i \, m_{i,t-1} + u_{i,t}$, *monthly mispricing levels. Half-life* $h^* = \ln(0.5)/\ln(\phi_i)$. *Newey–West HAC.*

Strategy	ϕ	HAC t	Half-life (months)
CDS–Bond Basis	0.846	(20.3)	**4.1**
BTP Italia	0.683	(10.6)	1.8
iTraxx Combined	0.569	(6.0)	1.2

- **CDS–Bond is the most persistent** (half-life $\approx$ 4 months): the only strategy that requires *continuous repo financing* of the cash bond on dealer balance sheets (Boyarchenko et al., 2016).
- BTP Italia is intermediate (half-life $\approx$ 2 months): retail buy-and-hold base, but mispricings are periodically refreshed by primary auctions reserved to households.
- iTraxx is the least persistent (half-life $\approx$ 1 month): a pure derivative basis with no continuous funding requirement; converges quickly across the dealer network.
- Persistence ranking is *not by liquidity of trading*, but by **funding-dependence of execution**.

P5: Common Factor → Intermediary Capital, *not* Macro

Factor	Coefficient	p-value
Panel A: Core Intermediary Proxies		$T = 149$, $\bar{R}^2 = 0.231$
HKM_IC	$-4.292**$	0.026
LIBOR_OIS	$+3.336***$	0.002
PB_EU_CDS_5Y	$-0.018**$	0.049
ILLIQ	$+1.730***$	0.005
Panel C: Macroeconomic Placebo		$T = 149$, $\bar{R}^2 = 0.112$
ΔUM	$+0.234$	0.984
ΔUR	$+11.027$	0.194
5Y5Y_INFL	-0.579	0.549
EPU_EU	$+0.003$	0.203

- **Panel A**: all four intermediary proxies significant with the predicted sign $\Rightarrow$ direct empirical content of Duffie (2010) and Prop. 6(i) of Brunnermeier and Pedersen (2009).
- **Panel C (placebo)**: *none* of the four macro proxies is significant; model not jointly significant (F-test $p = 0.23$) $\Rightarrow$ confirms common factor is balance-sheet, not macro.

P	Statement	Empirical evidence	
P1	Cross-market correlation	$\rho_{\Delta m} = +0.34 * **$ (CDS–Bond Basis vs iTraxx Combined)	✓
P2	Stress amplification	$\rho_{\mathrm{LOW}} = +0.10 \rightarrow \rho_{\mathrm{HIGH}} = +0.38$	✓
P3	Lead-lag transmission	BTP Italia $\bar{R}^2 = 13.3\%$ (iTraxx leads at 1%)	✓
P4	Persistence reflects funding-dependence	CDS–Bond Basis $\sim$4.1m ¿ BTP Italia $\sim$1.8m ¿ iTraxx Combined $\sim$1.2m	✓
P5	Common factor = intermediary, not macro	Intermediary $\bar{R}^2 = 23.1\%$ vs Macro placebo $\bar{R}^2 = 11.2\%$ (n.s.)	✓

Three reinforcing mechanisms:

- **Regulatory cost of intermediary capacity** (Boyarchenko et al., 2016):
  post-crisis SLR and Basel III have permanently raised the cost of dealer balance sheet, lifting the equilibrium level of basis trades — evident in the high persistence of CDS–Bond ($\phi = 0.85$, P4).
- **Slow-moving capital across strategies** (Duffie, 2010; Mitchell and Pulvino, 2012):
  capital reallocates across the dealer network on a horizon of several months. iTraxx adjusts first, BTP Italia last (P3), and the common factor is intermediary balance-sheet, not macro fundamentals (P5).
- **Investor-base segmentation as a structural feature**:
  BTP Italia retail bondholders generate a preferred-habitat (Vayanos and Vila, 2021) structurally insulated from institutional unwinding — the basis remains open for arbitrageurs who can deploy capital, but the funding-dependence of execution determines who can.

Bottom line: the geography of slow-moving capital is determined not by asset-class labels but by the network of intermediaries that service these markets — and ultimately by who holds the bonds.